Part-A Objective Questions

I. Write the correct option in the BOX only. [5*1 Mark = 5 Marks]

1. ***"Client requested software developers to add portability feature in the application"*** **Which one of the following software characteristics is client referring to?** A. Maintenance B. Transitional C. Operational D. Efficient	
2. ***"Software developers working on a new project want to avoid drawbacks of waterfall model. They want to test the application at each stage in reverse order."*** **Which process model would you recommend to the developers?** A. Waterfall model B. Spiral model C. V-model D. Agile model	
3. ***"Systems analyst needs to collect the requirements from the client in a way so that the client will get a chance to experience the features of their own application and provide a useful feedback to the analyst."*** **Which requirement elicitation technique should be considered by the analyst?** A. Oral interview B. Prototyping C. Task analysis D. Survey	
4. ***"System should ask for the user confirmation before deleting a file"*** **Which one of the following interface design components is the most suitable to fulfill the above requirement?** A. Dialogue box B. Application window C. Text-box D. Combo-box	
5. ***"Testing team wants to check whether the user-login functionality works correctly or not."*** **Which testing technique should be used by testing team?** A. Implementation testing B. Control flow testing C. White-box testing D. Black-box testing	

II. Identify the type of cohesion with the help of their definitions given in the left columns of the table below. **[5*0.5 Marks = 2.5 Marks]**

Select the most suitable type from the following types:
[Sequential, Communicational, Logical, Procedural, Functional, Temporal, Co-incidental]

Definition	Cohesion type
1. It is unplanned and random cohesion, which might be the result of breaking the program into smaller modules.	**Co-incidental**
2. Elements of module are organized such that they are processed at a similar point in time.	**Temporal**
3. Elements of module are grouped together, which are executed sequentially and work on same data.	**Communicational**
4. Elements of module are grouped because the output of one element serves as input to another module.	**Sequential**
5. Elements of module are grouped because they all contribute to a single well-defined function.	**Functional**

III. State True or False **[5*0.5 = 2.5 Mark]**

1. Organic projects are relatively small, simple software projects in which small teams with good application experience work together.	**True**
2. Validation is a process of examining whether or not the software satisfies the user requirements.	**True**
3. The greater the coupling, the better is the program design.	**False**
4. Modularization is a technique to divide a software system into multiple discrete and independent modules.	**True**
5. Embedded project are software projects in which teams with mixed experience levels must meet a mix of rigid and less than rigid requirements.	**False**

Part-B –Descriptive Answer Questions

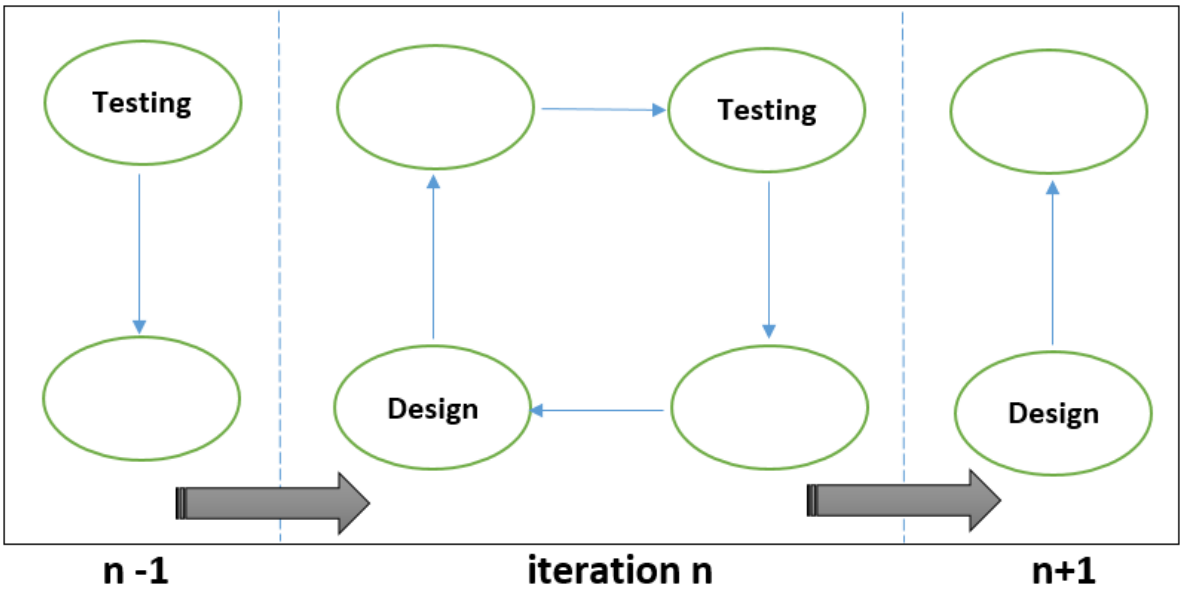
I. What do you understand by the term feasibility study as an important phase in SDLC?
[0.75 Mark]

At this step the team analyzes if software can be made to fulfill all requirements of the user and if there is any possibility of software being no more useful.

II. *"In software design, concurrency provides capability to the software to execute more than one part of code in parallel to each other.*
Give an appropriate example in support of the given statement. [0.75 Mark]

Example: The spell check feature in word processor is a module of software, which runs alongside the word processor itself.

III. Complete the iterative model diagram given below. [4*0.25=1 Mark]



IV. Differentiate between closed-ended interview and open-ended interview.
[0.75+0.75=1.5 Mark]

Structured (closed) interviews, where every single information to gather is decided in advance, they follow pattern and matter of discussion firmly.

Non-structured (open) interviews, where information to gather is not decided in advance, more flexible and less biased.

V. Write any two benefits of software validation and verification. [4*0.5=2 Mark]

Validation	Verification
1. It ensures the product under development is as per the user requirements. 2. It answers the question – "Are we developing the product which attempts all that user needs from this software?" 3. Validation emphasizes on user requirements.	1. It ensures the product being developed is according to design specifications. 2. It answers the question– "Are we developing this product by firmly following all design specifications?" 3. Verifications concentrates on the design and system specifications

VI. Explain any two testing levels in detail. [2*2 =4 Mark]

System Testing:

This can be accomplished using one or more of the following tests:
- Functionality testing-Tests all functionalities of the software against the requirement.
- Performance testing- This test proves how efficient the software is. It tests the effectiveness and average time taken by the software to do desired task.
- Security & Portability- These tests are done when the software is meant to work on various platforms and accessed by number of persons

Acceptance Testing:

This can be accomplished by the following two testing techniques:
- Alpha testing- The team of developer themselves perform alpha testing by using the system as if it is being used in work environment.
- Beta testing- After the software is tested internally, it is handed over to the users to use it under their production environment only for testing purpose.

Unit Testing

While coding, the programmer performs some tests on that unit of program to know if it is error free. Testing is performed under white-box testing approach. Unit testing helps developers decide that individual units of the program are working as per requirement and are error free.

Integration Testing
Even if the units of software are working fine individually, there is a need to find out if the units if integrated together would also work without errors. For example, argument passing and data updating etc.

Part C –Analytical Answer Questions

I. Software development team is working on a new application development project. An initial study has determined that the size of the program will be roughly 25,000 delivered source instructions (DSI). According to the project head, this project should be considered as Semi-detached software project.
Using the Basic COCOMO equations given below calculate a, b, c, and d. Show all the steps of your calculation.

Software project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

a) Effort applied in person month **[1.25 Marks]**

Effort (E) = ab * (KDSI)^{bb}

3.0 * (25)$^{1.12}$ = 110 Staff-Months

b) Time required to complete the project **[1.25 Marks]**

TDEV = *cb * (E)^{db}*

2.5 * (110)$^{0.35}$ = 13 Months

c) Average staff required **[1.25 Marks]**

Average Staffing: (E/TDEV)

110 staff-months / 13 months = 8 People

d) Productivity **[1.25 Marks]**

Productivity= DSI/Effort

25,000 /110 = 227 DSI/PM

II. Calculate the Function Point Measures for an average project with 8 inputs, 8 outputs, 3 inquiries, 4 master files and 5 interfaces. Estimate the size of the project considering 30 LOC per FP.

Complexity Levels	
Components	**Average**
Inputs	5
Outputs	5
Inquires	10
Master Files	15
Interfaces	10

a) Function Points: **[1 Mark]**

FP = (8 * 5) + (8 * 5) + (3 * 10) + (4 * 15) + (5 * 10)
FP = 40 + 40 + 30 + 60 + 50 FP = 220

b) Estimated Size: **[1 Mark]**

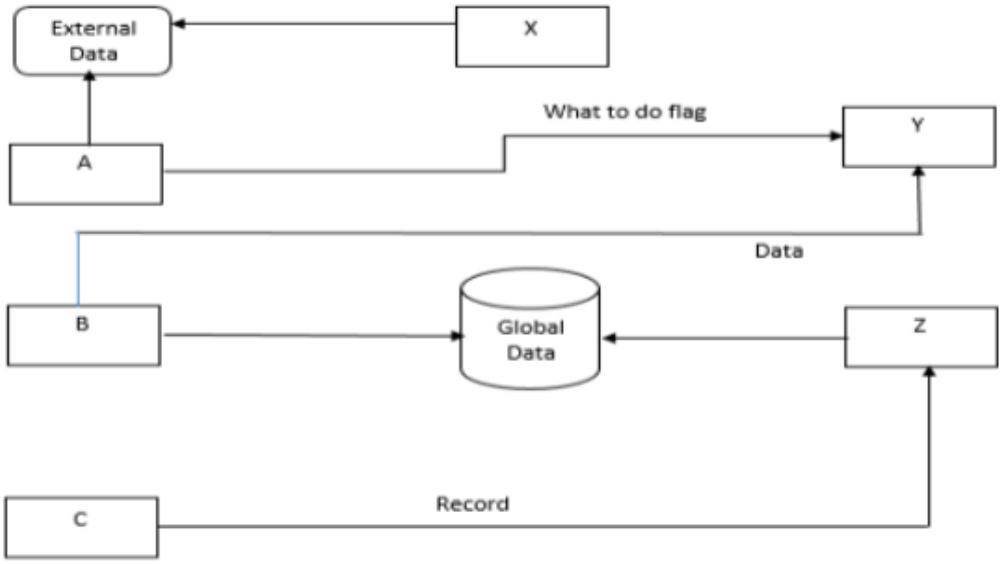
Size = FP * 50 Size = 220 * 30 Size = 6600 LOC

III. Create a test case for the login functionality do experience in college moodle (e-learn) system. Use the below give table and fill it with necessary details. **[5 Marks]**

Test Case	
Test Case ID: 1	**Test Title: Check Login in Moodle**
Test Case Description: This test case will test whether user is able to log in successfully by providing correct user id and password.	**Pre-Conditions: Login screen displayed**

Step	Test Steps	Expected Result	Actual Result	Status(Pass/Fail)	Remarks
1	Provide valid username and wrong password	Authentication fail	Authentication fail	Pass	Log in failed
2	Provide valid password and valid user name	Authorized	Authorized	Pass	Log in successfully

IV. Analyze the below given diagram and write the appropriate coupling types between Pairs of modules into the given table: [6*0.25=3 Marks]



Module	X	Y	Z
A	External coupling	Control coupling	No coupling
B	No coupling	Data coupling	Common/global coupling

V. Convert the following scenario into structure English. [5 Marks]

Criteria: If total purchase is more than or equal to 50 OMR, customer will get a discount of 25% on total bill amount. If total purchase is between 40 to 50 OMR, customer will get a discount of 15% on total bill amount. If total purchase is less than 40 OMR, no discount.

Solution:

If total purchase >= 50 OMR
    Discount = 25% on total bill
Else if total purchase between 40 and 50 OMR
    Discount = 15% on total bill
Else if total purchase <40 OMR
    Discount = NIL
End if
End if
End if